THE OPTIMAL CONTRACT MAKES A FLEXIBLE COMPUTING LOAD DECARBONIZE THE GRID: A SHAPE-INDEPENDENT GUARANTEE

TEAM A1 — ROLE 1 (OPTIMAL CONTRACT AND GENERAL CARBON THEORY)

Abstract. This is the Role 1 Week 3 deliverable, and it is a positive resolution of the Week-2 carbon question. Week 2 showed that under the *naive* dispatch rule—run the flexible computing load (FCL) on spare fossil whenever it pays—a finite interior optimum carries an irreducible fossil floor, so the carbon sign is ambiguous and, in general, not even decidable from the equilibrium cutoffs. Here we change the contract and the answer changes with it. We study the regulator-imposed *surplus contract*, which runs the FCL on renewable surplus and curtails it in fossil hours, and which minimizes the load's fossil footprint (driving it to zero). Three results follow, and all are positive. *First*, under the surplus contract the load burns *zero* fossil, so the Week-2 floor is gone by construction. *Second*—the keystone—whenever the load is adopted the contract strictly raises renewable investment, $K_R^c > K_R^\circ$, and total carbon *strictly falls*, $\mathbb{E}[\text{fossil}^c] < \mathbb{E}[\text{fossil}^\circ]$. The force of this is that, unlike the naive rule, it needs *no condition on the shape of* F: under the standing regularity (the renewable problem is single-peaked) and adoption, carbon falls whatever the tails or skew of demand. The carbon sign, which Week 2 showed is decided by the distributional shape under the naive rule, is freed from that shape by the contract. *Third*, the contract is a clean Pareto improvement on the no-load benchmark: profit strictly rises and carbon strictly falls, while reliability is held exactly at the no-load level (the surplus load never touches fossil, so it leaves the blackout target unchanged). The two standing hypotheses are adoption ($C_M < R\,F(0; K_R^\circ)$, the load is cheap enough to run on the available surplus) and the Week-2 single-peakedness; neither is a *sign-determining shape condition*—single-peakedness is the regularity that guarantees a well-defined optimum, not a restriction that decides which way carbon moves—and we are explicit that the surplus optimum may sit at the renewable-capacity bound when the surplus channel is strong. Every identity is checked against quadrature in `verify_week3.py`. The message is clean: a flexible computing load is a decarbonization instrument, and contract design is the lever that makes it one.

1. Setup: the model and the contract space

We keep the model and notation of the Week-1 and Week-2 deliverables. A regulated utility chooses renewable capacity K_R, dispatchable fossil K_F, and flexible-computing-load (FCL) capacity K_M. Residual demand is $\tilde{x} = \tilde{D} - K_R\tilde{S} \in [-K_R, \bar{D}]$ with CDF $F(x; K_R)$, density $f(x; K_R) > 0$ on $(-K_R, \bar{D})$, and $F_{K_R} := \partial F/\partial K_R > 0$ on the interior of the support (more renewables strictly shift residual demand toward surplus in every interior state—true for the scale model $\tilde{x} = \tilde{D} - K_R\tilde{S}$, where $F_{K_R} = \mathbb{E}[\tilde{S}\, f_{\tilde{D}}(x + K_R\tilde{S})] > 0$ wherever $f_{\tilde{D}} > 0$). This positivity is the only structural property of F we use, and it holds across surplus states ($x < 0$) and fossil states ($x > 0$) alike. Per-MWh: FCL revenue R, fossil cost c, price p, blackout penalty β; amortized capital costs C_R, C_F, C_M.

Assumption 1 (Regularity, as in Week 1)**.** $R > c + C_M$, $\quad R < c + C_F + C_M$, $\quad \beta + p > c + C_F$, $\beta + p > R$.

Dispatch and fossil accounting. In state x, consumers are served first (they outrank the FCL for scarce fossil), so consumer fossil is $\min\{(x)_+, K_F\}$. Renewable surplus $(-x)_+$ is free power. A *dispatch contract* is a measurable schedule $d : \mathbb{R} \to [0, K_M]$ with $d(x) \le (K_F - x)_+$ (the FCL can draw surplus plus fossil headroom); the fossil it burns is $(d(x) - (-x)_+)_+$ (surplus first, then fossil).

Date: Working results, Week 3.

Total fossil is

$$\mathbb{E}[\text{fossil}] = \underbrace{\mathbb{E}[\min\{(\tilde{x})_+, K_F\}]}_{\text{consumer fossil}} + \underbrace{\mathbb{E}[(d(\tilde{x}) - (-\tilde{x})_+)_+]}_{\text{FCL fossil}}. \tag{1}$$

Consumer fossil equals $\int_0^{K_F}(1-F)\,dx$ and does not depend on the contract d; only the second term does.

Two contracts. The Week-2 analysis used the *firm-optimal* (naive) contract $d^*(x) = \min\{(K_F - x)_+, K_M\}$, which runs the load whenever the marginal $R - c > 0$ is positive—hence on fossil headroom too. We contrast it with the *surplus contract*

$$d^c(x) := \min\{(-x)_+, K_M\}, \tag{2}$$

which runs the load only on renewable surplus and curtails it whenever fossil would be needed. We separate the two decision-makers cleanly: the *regulator* sets the real-time dispatch contract $d(\cdot)$ (the regulated utility dispatches as directed), and the *firm* then chooses capacities (K_R, K_F, K_M) to maximize profit under that contract. The surplus contract is regulator-imposed and is *not* dispatch incentive-compatible: with $R > c$ the firm would prefer to run the load on fossil headroom too (the naive d^*). It is enforced as a curtailment rule, exactly as demand-response and interruptibility tariffs are. The no-FCL benchmark (K_R°, K_F°) is the program with $K_M = 0$, with fossil $\mathbb{E}[\text{fossil}^\circ] = \int_0^{K_F^\circ}(1 - F(x; K_R^\circ))\,dx$ and the distribution-free cutoff

$$F(K_F^\circ; K_R^\circ) = \varphi^\circ := \frac{\beta + p - c - C_F}{\beta + p - c}. \tag{3}$$

2. The fossil-minimizing dispatch is surplus-only

We first show that, at any fixed capacities, the surplus contract minimizes the load's fossil footprint—driving it to zero—and that among zero-footprint contracts it is the one that monetizes the most surplus, hence the natural carbon-oriented contract. (We do *not* claim global carbon-optimality over the bilevel game: a different contract could induce a larger K_R at the price of some FCL fossil. Week 2 showed that trade—the naive rule buys a bigger capacity bump but pays the fossil floor—loses on net; the surplus contract takes the other side, zero footprint with a strict capacity gain, and §3 shows that side wins distribution-wide.)

Proposition 1 (Fossil-minimizing dispatch)**.** *Fix capacities (K_R, K_F, K_M). Among all admissible dispatch schedules d, total fossil (1) is minimized, state by state, by running the load on surplus alone, $d^c(x) = \min\{(-x)_+, K_M\}$, at which the FCL fossil term is identically zero. Any schedule that ever dispatches beyond the surplus, $d(x) > (-x)_+$, burns strictly more fossil in that state.*

Proof. Consumer fossil in (1) is independent of d, so minimizing total fossil is minimizing $\mathbb{E}[(d(\tilde{x}) - (-\tilde{x})_+)_+]$ pointwise. The integrand $(d(x) - (-x)_+)_+$ is zero whenever $d(x) \le (-x)_+$ and strictly increasing in $d(x)$ beyond that. Setting $d(x) = \min\{(-x)_+, K_M\} \le (-x)_+$ attains zero; any $d(x) > (-x)_+$ gives a strictly positive value. Hence d^c is the state-by-state fossil minimizer, with FCL fossil $\equiv 0$. Among the zero-fossil schedules ($d \le (-x)_+$), $d^c = \min\{(-x)_+, K_M\}$ is the one that runs the load to the full available surplus, so it maximizes FCL revenue $R\,\mathbb{E}[d]$ subject to zero fossil; it is therefore the zero-footprint contract the firm will actually operate (any $d < d^c$ earns less surplus revenue for the same zero footprint, and not operating at all, $d \equiv 0$, forgoes the revenue that makes adoption worthwhile). □

So the surplus contract is the one a regulator who wants to keep the load's benefits while erasing its fossil footprint would write: monetize the surplus, never burn fossil to mine. Under it, (1) collapses to consumer fossil,

$$\mathbb{E}[\text{fossil}^c] = \int_0^{K_F^c} (1 - F(x; K_R^c))\,dx, \tag{4}$$

exactly the no-FCL form—but evaluated at the capacities the contract induces, which is where the gain comes from.

3. The decarbonization theorem (no shape condition)

Under the surplus contract the firm's program is

$$\Pi^c = p\,\mathbb{E}[\tilde{D}] + R\int_{-K_M}^{0} F\,dx - c\int_0^{K_F}(1-F)\,dx - (\beta+p)\int_{K_F}^{\bar{D}}(1-F)\,dx - C_F K_F - C_R K_R - C_M K_M, \quad (5)$$

the FCL contributing the surplus revenue $R\,\mathbb{E}[\min\{(-\tilde{x})_+, K_M\}] = R\int_{-K_M}^{0} F\,dx$ and *no* fossil cost. Its first-order conditions are

$$\partial_{K_M}\Pi^c = R\,F(-K_M; K_R) - C_M = 0 \qquad \Longrightarrow \quad F(-K_M^c; K_R^c) = \tfrac{C_M}{R}, \quad (6)$$

$$\partial_{K_F}\Pi^c = (\beta+p-c)\big(1-F(K_F;K_R)\big) - C_F = 0 \qquad \Longrightarrow \quad F(K_F^c; K_R^c) = \varphi^\circ, \quad (7)$$

$$\partial_{K_R}\Pi^c = \underbrace{R\int_{-K_M}^{0} F_{K_R}\,dx}_{\geq 0,\ \text{surplus channel}} + c\int_0^{K_F} F_{K_R}\,dx + (\beta+p)\int_{K_F}^{\bar{D}} F_{K_R}\,dx - C_R = 0. \quad (8)$$

Two facts jump out. The fossil cutoff (7) is *identical* to the no-FCL one (3): $F(K_F^c; K_R^c) = \varphi^\circ = F(K_F^\circ; K_R^\circ)$. And the renewable condition (8) is the no-FCL renewable condition plus the strictly positive *surplus channel* term $R\int_{-K_M}^{0} F_{K_R}\,dx$. The FCL is adopted ($K_M^c > 0$) when (6) has an interior root; a sufficient condition (using $F(0;K_R^c) \geq F(0;K_R^\circ)$, since $K_R^c \geq K_R^\circ$ and $F(0;\cdot)$ is increasing) is

$$C_M < R\,F(0;K_R^\circ) \qquad \text{(adoption: the load is cheap enough to run on the surplus).} \quad (9)$$

Condition 1 (Single-peaked renewable problem)**.** The renewable marginal value (the left side of (8) plus C_R) is strictly decreasing in K_R along the locus (7). This is the Week-2 reduced second-order condition (the Schur/diagonal-dominance condition), here for the surplus program; it holds on the regular region and at the power-law boundary it is the knife-edge.

The decarbonization theorem rests on a single monotone fact.

Lemma 1 (More renewables, less fossil)**.** *Along the fossil-cutoff locus $K_F(K_R) = F^{-1}(\varphi^\circ; K_R)$, the utility's own fossil output $e(K_R) := \int_0^{(K_F(K_R))_+}(1-F(x;K_R))\,dx$ is non-increasing in K_R, and strictly decreasing wherever $K_F(K_R) > 0$:*

$$e'(K_R) = \underbrace{(1-\varphi^\circ)\,K_F'(K_R)}_{\leq 0} - \underbrace{\int_0^{K_F(K_R)} F_{K_R}(x;K_R)\,dx}_{>0} \;<\; 0, \qquad K_F'(K_R) = -\frac{F_{K_R}(K_F;K_R)}{f(K_F;K_R)} \leq 0.$$

Proof. Differentiate e by Leibniz, using $1 - F(K_F;K_R) = 1-\varphi^\circ$ at the moving upper limit; K_F' is the implicit derivative of $F(K_F(K_R);K_R) = \varphi^\circ$. Both terms are ≤ 0, the integral strictly so by $F_{K_R} > 0$: building renewables both lowers the dispatchable-fossil cutoff and displaces fossil inside the served band. Where $K_F(K_R) \leq 0$ the integral is empty and $e \equiv 0$, so e is non-increasing throughout. □

Theorem 1 (Decarbonization, with no sign-determining shape condition)**.** *Under Condition 1, the surplus contract satisfies*

$$\boxed{K_R^c \geq K_R^\circ \quad \textit{and} \quad \mathbb{E}[\text{fossil}^c] \leq \mathbb{E}[\text{fossil}^\circ],}$$

both with strict *inequality exactly when the load is adopted* (9)*. The carbon inequality is an immediate consequence of the renewable one:* more renewables mean less fossil*. The conclusion uses only the standing positivity $F_{K_R} > 0$ on the interior support and the regularity Condition 1 (which guarantees a well-defined optimum and can fail, e.g. at the power-law knife-edge); given an optimum and adoption, the* sign *of the carbon change carries* no *dependence on the shape, tails, or skew of F. This is the sharp contrast with Week 2, where (Remark 1) the conditional shape of F decided the sign; the surplus contract removes that shape-dependence. (Magnitude still depends on shape; only the sign is freed.)*

Proof. Renewables ($K_R^c \geq K_R^\circ$). If the load is not adopted ($K_M^c = 0$) the surplus program is the no-FCL program and $K_R^c = K_R^\circ$. If it is adopted, evaluate the renewable marginal value (8) at (K_R°, K_F°): by (7) the fossil cutoff there is K_F°, so its last two integrals are the no-FCL renewable benefit, which equals C_R at K_R°. The remaining surplus-channel term is strictly positive,

$$\partial_{K_R}\Pi^c\big|_{(K_R^\circ, K_F^\circ)} = R\int_{-K_M}^{0} F_{K_R}(x; K_R^\circ)\,dx > 0 \qquad (F_{K_R} > 0 \text{ on } (-K_M, 0)),$$

so the renewable marginal value exceeds C_R at K_R°; by Condition 1 it is strictly decreasing, so the surplus optimum—its zero if interior, otherwise the capacity bound $\bar{K}_R$—satisfies $K_R^c > K_R^\circ$.

Carbon ($\mathbb{E}[\text{fossil}^c] \leq \mathbb{E}[\text{fossil}^\circ]$). Under the surplus contract the load burns no fossil (Prop. 1), so total fossil is exactly the utility's own output $e(K_R^c)$, while the benchmark is $e(K_R^\circ)$. By Lemma 1 e is non-increasing and strictly decreasing while $K_F > 0$; with $K_R^c \geq K_R^\circ$ this gives $e(K_R^c) \leq e(K_R^\circ)$, strict whenever $K_R^c > K_R^\circ$, i.e. whenever the load is adopted. That is the whole argument: the contract zeroes the load's fossil and weakly raises renewables, and more renewables mean less fossil.[1] □

Remark 1 (Why this resolves Week 2). Week 2 proved that under the naive contract the carbon sign is *irreducibly distributional*: at fixed equilibrium capacities the admissible conditional laws make Δ range over an interval straddling zero, so no distribution-free statement is possible and a fossil floor $(K_M - K_R)_+$ blocks reduction at interior optima. Theorem 1 shows the obstruction was an artifact of the *contract*, not the technology. The surplus contract sets the FCL fossil—and with it the floor—to zero, and the only remaining channel is the renewable build-out, which has a definite sign. The carbon question goes from "it depends on the tails of F" to "it always falls when the load is adopted."

4. The Pareto improvement and the exact saving

Corollary 1 (The surplus contract Pareto-dominates no load)**.** *Under the hypotheses of Theorem 1, relative to the no-FCL benchmark the surplus contract* (i) *strictly raises the utility's profit,* (ii) *holds the blackout probability* exactly *at the no-load level, and* (iii) *strictly lowers carbon. It is thus a strict Pareto improvement: more profit and less carbon at no reliability cost. The contract is regulator-imposed, not firm-optimal: left to itself the firm would also run the load on fossil headroom (the naive rule) and earn still more. But it strictly prefers operating under the surplus contract to having no load, so participation is individually rational and no subsidy is needed—the regulator gives up only the firm's extra* fossil-powered *computing profit, which is exactly the carbon-causing margin.*

Proof. (iii) is Theorem 1. (i): the surplus contract adds the option of earning R on otherwise-curtailed surplus at hardware cost C_M; by (9) the first FCL unit's marginal surplus revenue $R\,F(0; K_R^\circ)$ exceeds C_M, so $\Pi^c > \Pi^\circ$ strictly (the firm can always replicate the benchmark by setting $K_M = 0$, and adoption is profitable). (ii): blackout occurs on $\{\tilde{x} > K_F\}$ with probability $1 - F(K_F; K_R)$. By the

[1] Verified in `verify_week3.py`: across symmetric, left- and right-skewed, bimodal, and high-capacity-factor shapes, the surplus-channel term is strictly positive at K_R°, $K_R^c > K_R^\circ$, and $\mathbb{E}[\text{fossil}^c] < \mathbb{E}[\text{fossil}^\circ]$ at every adopted optimum—the sign never flips.

fossil cutoff (7), *both* programs target the same reliability, $F(K_F^c; K_R^c) = \varphi^\circ = F(K_F^\circ; K_R^\circ)$, so the blackout probability is $1 - \varphi^\circ$ in both—identical. The surplus load never uses fossil, so it leaves the fossil-capacity trade-off, and hence the blackout target, untouched; the larger K_R^c is spent on *lowering fossil capacity* ($K_F^c \le K_F^\circ$, since the fossil cutoff falls in K_R by Lemma 1) rather than on extra reliability. Reliability is therefore held, not improved—a clean tie, which is what makes the result a Pareto improvement.[2] □

Proposition 2 (Exact carbon saving and its monotonicity)**.** *The carbon saving under the surplus contract is*

$$\Delta^c := \mathbb{E}[\text{fossil}^\circ] - \mathbb{E}[\text{fossil}^c] = \int_0^{K_F^c} [F(x; K_R^c) - F(x; K_R^\circ)]\, dx + \int_{K_F^c}^{K_F^\circ} (1 - F(x; K_R^\circ))\, dx \;>\; 0. \quad (10)$$

It is increasing in the surplus channel: a larger adopted K_M^c (cheaper load, more surplus) raises K_R^c and hence Δ^c. In the cheap-renewable limit $K_R^c \to \infty$ the contract drives the utility to full decarbonization, $\mathbb{E}[\text{fossil}^c] \to 0$*, so* $\Delta^c \to \mathbb{E}[\text{fossil}^\circ]$.

Proof. In the interior case $K_F^c > 0$ the identity is the expansion of $e(K_R^\circ) - e(K_R^c)$ used in Theorem 1. Monotonicity is Lemma 1: $\mathbb{E}[\text{fossil}^c] = e(K_R^c)$ is strictly decreasing in K_R^c, and a larger adopted K_M^c enlarges the surplus channel and raises K_R^c, so $\Delta^c = \mathbb{E}[\text{fossil}^\circ] - e(K_R^c)$ strictly increases. As $K_R^c \to \infty$, $F(x; K_R^c) \to 1$ for each fixed $x > 0$ (renewables swamp demand), so by dominated convergence $e(K_R^c) \to 0$ and $\Delta^c \to \mathbb{E}[\text{fossil}^\circ]$ (full decarbonization). □

Remark 2 (The contract incentivizes renewable build-out). Proposition 2 says the surplus contract does not merely fail to add fossil—it actively pulls the optimal renewable capacity up, and in the cheap-renewable regime all the way to full decarbonization. The flexible load, correctly contracted, behaves like a subsidy for renewables financed by selling surplus power to computing. This is the positive mechanism the paper has been after: not "computing happens to be green," but "the right contract makes computing pay for the green build-out."

5. Numerical validation

All claims are checked in `verify_week3.py` against adaptive quadrature of the profit integrals, on non-power-law Beta scale mixtures $\tilde{x} = \tilde{D} - K_R \tilde{S}$ spanning symmetric, left-skewed, right-skewed, and bimodal demand. The suite confirms, at every adopted optimum and across the family: the surplus contract burns zero FCL fossil (Prop. 1); the surplus-channel term in (8) is strictly positive at K_R°, giving $K_R^c > K_R^\circ$; the fossil output $e(K_R)$ is strictly decreasing (Lemma 1); and the strict carbon fall $\mathbb{E}[\text{fossil}^c] < \mathbb{E}[\text{fossil}^\circ]$ holds with the *same sign for every shape in the family* (Theorem 1); the Pareto improvement $\Pi^c > \Pi^\circ$ with blackout held equal, blackout^c =blackout$^\circ$ $= 1 - \varphi^\circ$ (Cor. 1); and the exact saving (10) matches the direct quadrature. As a contrast the suite also reports the Week-2 naive-rule fossil at the same primitives, exhibiting the floor that the contract removes. The suite reports a single pass count.

[2] Verified in `verify_week3.py`: $\Pi^c > \Pi^\circ$ and blackout^c =blackout$^\circ$ (to numerical tolerance) at every adopted optimum.